

HOSPITAL QUEUE SIMULATION AND MODELLING SYSTEM

1. PROPOSED TITLE:

Simulation-Based Optimization of Hospital Queue Management

2. Introduction, Problem statement, General and specific objectives:

- Introduction: The project aims to improve patient flow in hospital queues using simulation.
- Problem statement: Inefficient queue management leads to long waiting times and reduced hospital performance.
- General objective: Optimize hospital queue processes to enhance service delivery.
- Specific objectives:
 - i. Model current queue behavior;
 - ii. Design an improved queue system;
 - iii. Validate the model against real performance indicators.

3. The current working system:

Existing hospital queue processes rely on manual scheduling and basic performance tracking, analyzed with Python/Excel for data insights and documented in Microsoft Word.

4. Proposed system:

A simulation model that predicts and optimizes patient flow, validated by comparing outputs with observed hospital performance indicators.

5. Functional requirements:

- Data collection and analysis (Python/Excel).
- Simulation modeling of queue dynamics.
- Reporting and documentation in Microsoft Word.

6. Non-functional requirements

- Anonymization of patient data for ethical compliance.
- Academic-only use of data.
- Timely and accurate simulation results.

7. Methodology

- Data analysis using Python/Excel to understand current performance.
- Development of a simulation model for queue management.
- Model validation against actual hospital performance indicators
- Ethical handling of patient data through anonymization